International Journal of Mechanical Sciences 269 (2024) 109080



Contents lists available at ScienceDirect

International Journal of Mechanical Sciences

journal homepage: www.elsevier.com/locate/ijmecsci





A physics-informed neural network for Kresling origami structures

Chen-Xu Liu [a], Xinghao Wang [a], Weiming Liu [b,*], Yi-Fan Yang [c], Gui-Lan Yu [d], Zhanli Liu [a,*]

[a] Applied Mechanics Lab., Department of Engineering Mechanics, School of Aerospace, Tsinghua University, Beijing 100084, China
[b] Department of Critical Care Medicine, China Rehabilitation Research Center, Capital Medical University, Beijing 100068, China
[c] China Academy of Building Research, Beijing 100013, China
[d] School of Civil Engineering, Beijing Jiaotong University, Beijing 100044, China

ARTICLE INFO

Keywords:
Origami structure
Machine learning
Potential energy
Kresling pattern
Physics-informed neural network
Prediction and design

ABSTRACT

Origami structures have the advantages of foldability and adjustability, with applications spanning numerous engineering fields. However, there remains a dearth of intelligent and convenient methods that can effectively tackle both potential energy prediction and design problems on origami structures. This study proposes a novel physics-informed neural network (PINN) to predict and design potential energy curves of Kresling origami structures without labelled data. A sorting operation is coupled into the PINN, ensuring the prediction correctness. The accuracy of the potential energy curves predicted by the PINN is demonstrated through comparison with a reference and the exhaustive method. A prediction only takes less than one second and the precision of the PINN significantly surpasses that of the exhaustive method, proving the extremely high efficiency and credibility of the PINN. Furthermore, two design cases for Kresling origami structures, matching a target potential energy curve and a set of target potential energy points, are performed. The designed structures meet the expectations and each design takes a few seconds, showing the efficiency and applicability of the PINN in inverse design. The presented physics-driven approach without labelled data offers an innovative tool with learning ability to predict and design. It also provides a valuable reference for the force and stiffness design of Kresling origami structures. In addition, the code of the PINN is shared online.

1. Introduction

Origami, a traditional art of paper folding, has transcended its ancient roots to become a source of inspiration for modern engineering [1–3]. The delicate and intricate folds that once adorned decorative paper have given rise to a remarkable and innovative field known as origami structures [4–6]. These structures, as a fusion of art and science, have evolved into a potent tool with applications in various domains, such as aerospace [7–9], robotics [10–12], architecture [13–15], and biology [16–18], due to their foldability, adjustability, and lightweight.

Common origami patterns include Kresling [19–21], Waterbomb [22,23], Miura [24–26], and Tachi-Miura Polyhedron [27,28]. Recently, numerous scholars have proposed innovative structures inspired by these origami patterns and subjected them to analyses of their mechanical performances, showcasing their engineering viability [29–31]. Research on robots drew many valuable references from origami structures with multi-stability and foldability [32–35]. Since origami structures have good deformability, lots of energy-absorbing and impact-resistant structures were inspired by them [36–39]. To enhance the load-bearing capacity of origami structures, many types of origami patterns were ameliorated, increasing the stiffness [40–44]. Due to the nonlinear deformation, origami structures possess quasi-zero stiffness properties, so they can serve as good low-frequency vibration isolators [45–49]. Furthermore, origami structures with quasi-zero stiffness properties can be used to form phononic metamaterials, realizing ultra-low-frequency wave control [50–55].Nevertheless, the development of these structures predominantly relies on human knowledge and experience, and it is uneasy to achieve desired mechanical performances.

Optimization algorithms can be used to improve structural performances [56–58], so some design approaches have been applied to origami structures. Dang et al. [59] and Wang et al. [60] respectively developed a sequential two-stage optimization framework and an optimality criteria-based algorithm to optimize origami structures by minimizing the strain energy, which aim to approximate targeted origami shapes. A level set method was adopted to assist the generative design of origami structures, meeting the kinematic requirements [61]. Sareh [62] used concepts from geometry and crystallography to address

* Corresponding authors.
E-mail addresses: liuweiming1115@163.com (W. Liu), liuzhanli@mail.tsinghua.edu.cn (Z. Liu).

https://doi.org/10.1016/j.ijmecsci.2024.109080
Received 11 November 2023; Received in revised form 25 January 2024; Accepted 31 January 2024
Available online 4 February 2024
0020-7403/© 2024 Elsevier Ltd. All rights reserved.

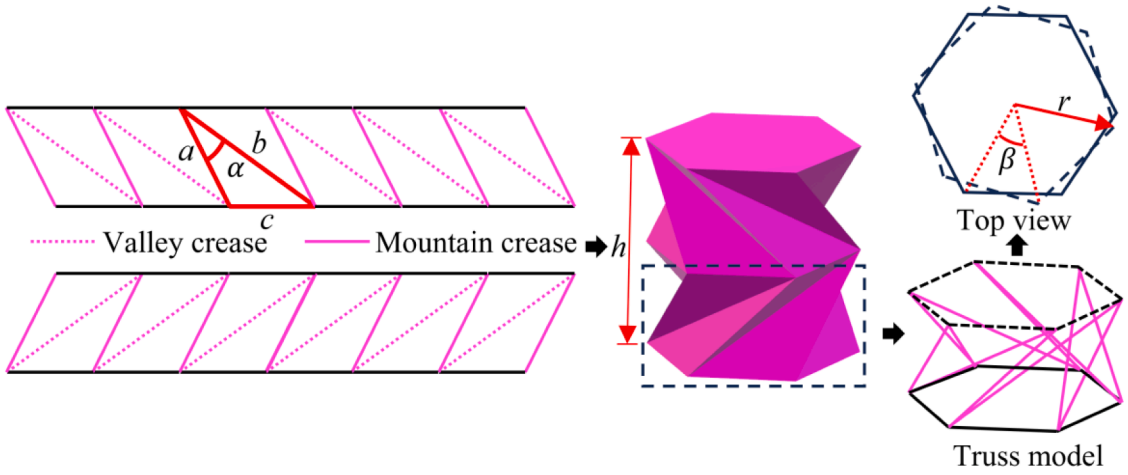


Fig. 1. Kresling origami structure with mirror symmetry; *a, b*, and *c* are the undeformed side lengths of fold surfaces; α is the angle between *a* and *b; h* is the height between the bottom and top planes; *r* is the outer circle radius of polygons; β is the rotation angle of the middle polygon.

the design problem of the least symmetric derivative of Miura origami. The combination of a particle swarm optimization algorithm and a semi-analytical periodic spectral plate method [63] or a graphical method [64,65] was established to optimize origami structures. However, more efficient and intelligent design methods with learning ability remain a significant challenge for origami structures.

Machine learning has opened up a promising avenue with efficiency and intelligence in science and engineering [66]. It has been instrumental in addressing a wide array of challenges across various domains, including medicine, chemistry, and mechanics [67–69]. In the field of origami structures, Zhang et al. [70] harnessed data-driven machine learning models for predicting structural mechanical behaviors. Zhu and Filipov [71] adopted the same way to design origami shapes. Moshtaghzadeh et al. [72] trained an artificial neural network with finite element-obtained data to assess fatigue and stability of an origami-inspired structure. However, both approaches come with the inconvenience of requiring extensive datasets. Furthermore, the performance of these machine learning models is inherently limited to the generalization domain covered by the data, which hampers the exploration of novel structures. Hence, there is a pressing need for the development of novel machine learning models tailored for origami structures to overcome the aforementioned limitations of the data-driven approach.

In this study, we propose a novel physics-informed neural network (PINN) without labelled data to predict and design potential energy curves of Kresling origami structures. The PINN combines a neural network with the governing equations of origami structures. This fusion empowers the PINN to provide predictions and designs without relying on training samples, working effectively within a broad range solely constrained by the governing equations. An advanced improvement is coupled into the PINN to ensure the prediction correctness, which is a sorting operation in the last layer of the neural network. Constraints on design variables are enforced during the PINN designing since these variables are bounded. For the prediction and design problems, two loss functions with physical connotation are defined. To assess the prediction reliability of the PINN, the potential energy curves predicted by the PINN are compared with those from a reference and by the exhaustive method. The effect and necessity of the sorting operation are proved by analyzing the PINNs with or without the advanced improvement. To demonstrate the design capability of the PINN, two design cases with different target forms are performed. This study also provides a valuable reference for the force and stiffness design of Kresling origami structures since both of them are related to the first and second derivatives of potential energy, respectively.

The remaining sections of this paper are structured as follows: Section 2 introduces Kresling origami structures, Section 3 describes the PINN for the potential energy prediction and inverse design of Kresling origami structures, Section 4 provides results and discussions, and Section 5 presents the conclusion.

2. Kresling origami structures

The considered Kresling origami structures are introduced in this section. As shown in Fig. 1, Kresling pattern is a common type of origami, which can be obtained by folding a 2D flat sheet. It is widely used in various structures, such as robotic arms [73], vibration isolators [74], and shock absorbers [75]. In this study, the structure with two mirror symmetry Kresling patterns is considered, eliminating the rotation of the bottom and top planes [76]. The material is assumed as perfect elastic medium and the Kresling origami structure is made up of flat sheets, thus this structure can be equivalent to the truss model for the convenience of investigation, where each side is replaced by spring rods along their creases [77–79]. The undeformed lengths of each side are represented as *a, b*, and *c*, where *c* is the bottom side and *a* is the shortest side of the remaining two sides. The angle between *a* and *b* is denoted as α. Additionally, the side number of the polygon on the bottom/middle/top plane is *n*, and the outer circle radius of these polygons is labelled as *r*. Through geometric derivation, a relationship among the variables mentioned is established, which is as follows

$$\begin{cases} c = 2r\sin\left(\dfrac{\pi}{n}\right), \\ b = \dfrac{c\sin\left(\alpha + \arcsin\left(\dfrac{a\sin(\alpha)}{c}\right)\right)}{\sin(\alpha)}. \end{cases} \tag{1}$$

According to Eq. (1), the shape of the Kresling origami structure can be determined by *a, α, n*, and *r*. Here, only the displacement along the direction perpendicular to the polygon plane is taken into account. As a result, only the lengths *a* and *b* will change over the distance between the bottom and top planes, which is denoted as height *h*.

For the generality, *a* and *h* are made dimensionless using the following equations

$$\overline{a} = \frac{a}{r} \tag{2a}$$

$$\overline{h} = \frac{h}{r}. \tag{2b}$$

The post-deformation dimensionless lengths of spring rods *a* and *b* can be respectively expressed as

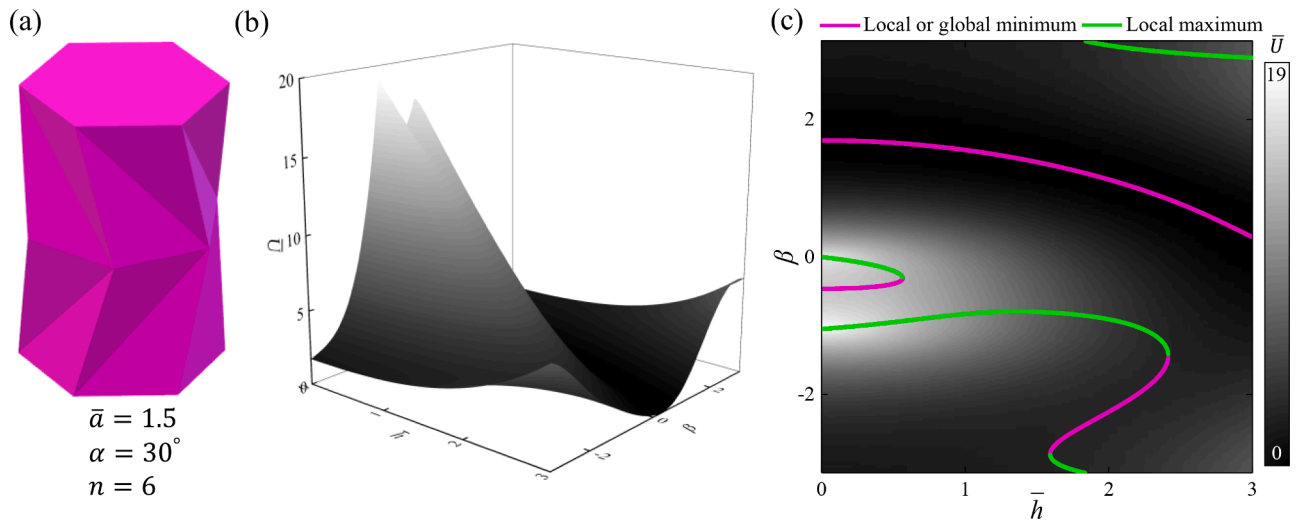


Fig. 2. The variations of the dimensionless potential energy $\overline{U}$ with the dimensionless height of Kresling origami structure $\bar{h}$ and the rotation angle of the middle polygon β; (a) the Kresling origami structure, where $\bar{a} = 1.5$, $\alpha = \pi/6$, and $n = 6$; (b) 3D cloud map; (b) 2D cloud map, where the purple (local or global minimum for each $\bar{h}$) and green (local maximum for each $\bar{h}$) solid lines are the solutions meeting Eq. (6).

$$\bar{a}_{\mathrm{d}} = \sqrt{\frac{\bar{h}^2}{4} + 4\sin^2\left(\frac{\beta}{2}\right)} = \frac{a_{\mathrm{d}}}{r} \tag{3}$$

$$\bar{b}_{\mathrm{d}} = \sqrt{\frac{\bar{h}^2}{4} + 4\sin^2\left(\frac{\beta}{2} + \frac{\pi}{n}\right)} = \frac{b_{\mathrm{d}}}{r} \tag{4}$$

where β is the rotation angle of the middle polygon, where $\beta = 0$ when $\bar{a}_{\mathrm{d}}$ is equal to $\bar{h}/2$.

According to the deformation of spring rods, the dimensionless potential energy of the Kresling origami structure can be written as follows

$$\overline{U} = n\left((\bar{a}_{\mathrm{d}} - \bar{a})^2 + \eta_k(\bar{b}_{\mathrm{d}} - \bar{b})^2\right) = \frac{U}{k_a r^2} \tag{5}$$

where $\eta_k = k_b/k_a = a/b$; k_a and k_b are the stiffnesses of spring rods a and b, respectively; the structure weight is ignored to simplify the problem. If multiple same Kresling origami structures are arranged in series or parallel, Eq. (5) multiplied by the number of these Kresling origami structures can be used as the potential energy of this structural system [76]. The force acting in the height direction is uniformly applied to the top polygon, leading to a situation where the torque in the middle polygon is zero, namely $\partial\overline{U}/\partial\beta = 0$ [76]. As a result, a constraint condition relating h and β can be derived, which is as follows

$$\left(1 - \frac{\bar{a}}{\bar{a}_{\mathrm{d}}}\right)\sin(\beta) + \eta_k\left(1 - \frac{\bar{b}}{\bar{b}_{\mathrm{d}}}\right)\sin\left(\beta + \frac{2\pi}{n}\right) = 0. \tag{6}$$

Obviously, Eq. (6) is a complex transcendental equation, which contains Eqs. (3) and (4). Fig. 2 shows the solutions of potential energy meeting Eq. (6) of a Kresling origami structure, which is denoted by purple and green solid lines, where $\bar{a} = 1.5$, $\alpha = \pi/6$, and $n = 6$. It can be seen that there exist plural solutions for a specified $\bar{h}$ even when considering the second derivative of $\overline{U}$ with respect to β. However, only

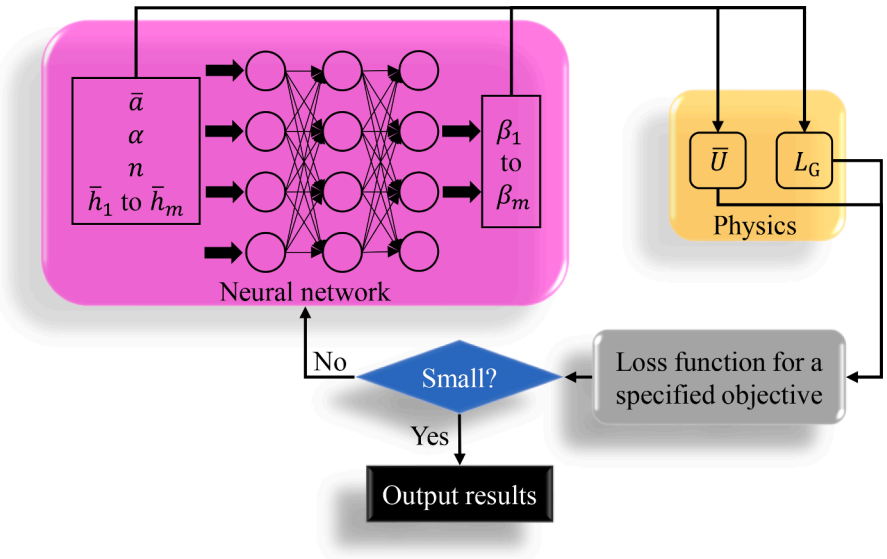


Fig. 3. The architecture of the physics-informed neural network (PINN) for the potential energy prediction and inverse design of Kresling origami structures; $\bar{a}$, α, and n are input geometry parameters, which determine the shape of a Kresling origami structure; $\bar{\boldsymbol{h}} = [\bar{h}_1, \bar{h}_2, \ldots, \bar{h}_m]$ is the input list vector containing different height states, where m is the dimension of the vector $\bar{\boldsymbol{h}}$; $\boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_m]$ is the output list vector containing rotation angle states of the middle polygon that $\bar{\boldsymbol{h}}$ corresponds to; $\overline{U}$ is Eq. (5), which represents the potential energy; L_{G} is Eq. (8), which makes the output $\boldsymbol{\beta}$ meet the physical constraint.

the global minimum for each $\overline{h}$ is the desired potential energy solution since prestress is not applied to the Kresling origami structure in this study. The existence of plural solutions in Eq. (6) poses a challenge for both forward prediction and inverse design of Kresling origami structures.

3. Physics-informed neural network

There are a lot of methods to solve the transcendental equation Eq. (6), such as exhaustive method, dichotomy, and Newton-Raphson method. However, it is troublesome for these methods to identify the desired relation of potential energy and height, and it is difficult to achieve inverse designs solely through themselves. The data-driven machine learning method holds promise for achieving accurate forward prediction and inverse design of origami structures [80]. However, it demands a substantial amount of pre-existing solved data, which is a huge amount of work. Furthermore, it operates within a limited working domain, constraining its applicability. To address these problems, a new method needs to be explored.

Relying on priori knowledge and powerful simulation capability, PINNs, a fusion of machine learning and physical equations, can tackle the prediction and design problems of structures even without labelled data [81,82]. Through defining a reasonable loss function, PINNs can obtain the desired solution. They also have the ability of designing structures by backpropagation since they are differentiable and continuous functions. In this study, a PINN tailored for Kresling origami structures is proposed to achieve the potential energy prediction and inverse design. The architecture, the advanced improvement, and the modifications for inverse design of the PINN are presented in the following sections.

3.1. Model architecture

The PINN for Kresling origami structures is mainly composed of a neural network, equations with physical information, and a loss function for a specified objective (such as prediction and design), as shown in Fig. 3. The inputs for the neural network include a vector composed of $\overline{h}$ values distributed within a range and feature parameters determining the shape of a Kresling origami structure, which contain $\overline{a}$, α, and n. According to Eq. (5), the potential energy $\overline{U}$ can be calculated based on $\overline{h}$ and β in the situation where shape feature parameters are known. Hence, a vector composed of β values corresponding to $\overline{h}$ is taken as the output. The neural network has three layers, where each of the first two layers has 128 neurons and the neuron number of the last layer is determined by the dimension of the input vector $\overline{\boldsymbol{h}}$. The activation function of the first two layers is set to be Sigmoid, enhancing the nonlinear simulation capability of the neural network. Considering that the rotation range of the middle polygon is difficultly beyond the range from $-\pi$ to π, the activation function of the last layer is set as follows

$$\beta = \pi \sin(z) \tag{7}$$

where z is the pre-activated value of the last layer.

In order to make the output β meet the physical constraint, the PINN contains the following equation based on Eq. (6), which is expressed as

$$L_{\mathrm{G}} = \left(\left(1-\frac{\overline{a}}{\overline{a}_{\mathrm{d}}}\right)\sin(\beta) + \eta_k\left(1-\frac{\overline{b}}{\overline{b}_{\mathrm{d}}}\right)\sin\left(\beta+\frac{2\pi}{n}\right)\right)^2 \tag{8}$$

where the smaller the value of L_{G}, the better the output β satisfies the governing Eq. (6). Given that both prediction and design problems are related to potential energy, Eq. (5) is integrated into the PINN.

For different objectives, the loss function needs to be tailored. Considering that there is no prestress in the Kresling origami structure, the potential energy $\overline{U}$ is always the smallest for each $\overline{h}$ according to the minimum total potential energy principle. Hence, the loss function for potential energy prediction can be expressed as follows

$$L_{\mathrm{P}} = \frac{1}{m}\sum_{i=1}^{m}\overline{U}^{(i)} \tag{9}$$

where m is the dimension of the input vector $\overline{\boldsymbol{h}}$ or the output vector $\boldsymbol{\beta}$. To meet a specified target potential energy curve, the loss function for the inverse design can be defined as follows

$$L_{\mathrm{D}} = \frac{1}{m}\sum_{i=1}^{m}\left(\lambda_1\left(\overline{U}^{(i)} - \overline{U}_{\mathrm{T}}^{(i)}\right)^2 + \lambda_2 L_{\mathrm{G}}^{(i)}\right) \tag{10}$$

where $\overline{U}_{\mathrm{T}}$ is the target potential energy; λ_1 and λ_2 are the weights of the loss terms, both of which are set to be 1.

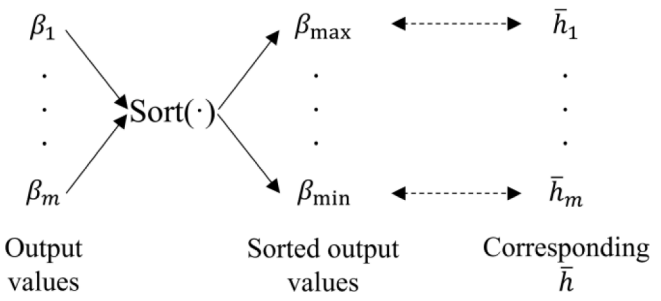


Fig. 4. The sorting operation in the last layer of the neural network; $\overline{h}_1, \ldots, \overline{h}_m$ are different height states of the Kresling origami structure; $\beta_1, \ldots, \beta_m$ are outputs of the neural network, which represent different rotation angle states of the middle polygon; $\beta_{\max}, \ldots, \beta_{\min}$ are the sorted results of $\beta_1, \ldots, \beta_m$ from largest to smallest, which correspond to $\overline{h}_1, \ldots, \overline{h}_m$, respectively.

3.2. An advanced improvement of the model

As illustrated in Fig. 2, there may be plural β values satisfying the governing Eq. (6) for each $\overline{h}$, but there is only one meeting the expectation. Additionally, since the middle polygon will rotate counterclockwise during compression where a is a shorter value between the two sides except for c in a fold surface, β is monotonically decreasing with the increase of $\overline{h}$. Hence, to ensure the correctness of the output as much as possible, the output values of the last layer are sorted from largest to smallest, as shown in Fig. 4. This operation is equivalent to imposing an additional constraint on the neural network, reducing the ineffective search space.

3.3. Modifications for inverse design

The PINN possesses the capability not only to predict potential energy but also to design Kresling origami structures, owing to its inherent nature as a differentiable and continuous function. However, limiting the ranges of the design variables $\overline{a}$ and α is essential to ensure they adhere to the geometrical constraints of Kresling origami structures. Failure to do so may lead to training divergence when using the PINN for design purposes. Given that α must be within the range from 0 to π and $\overline{a}$ must be within the range from 0 to $\overline{c}/\sin(\alpha)$ where $\overline{c} = c/r$, two equations are established in the PINN to limit the ranges of the design variables, as shown in Fig. 5, which are as follows

$$\alpha = \frac{\pi}{1+\mathrm{e}^{-\delta_\alpha}} \tag{11}$$

$$\overline{a} = \frac{\overline{c}}{(1+\mathrm{e}^{-\delta_{\overline{a}}})\sin(\alpha)} \tag{12}$$

where δ_α and $\delta_{\overline{a}}$ are pre-input or pre-activated variables for α and $\overline{a}$, respectively, which are initially set to be zero when designing.

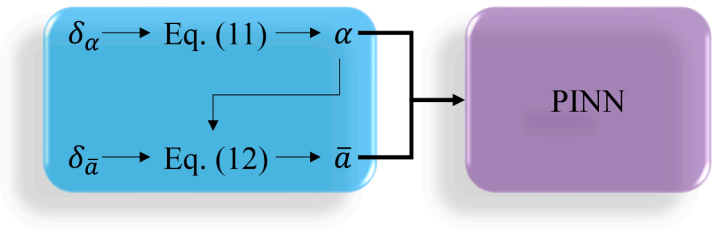


Fig. 5. Modifications of the PINN for inverse design; α and $\bar{a}$ are design variables; δ_α and $\delta_{\bar{a}}$ are pre-activated variables for α and $\bar{a}$, respectively; Eqs. (11) and (12) are defined bounded activation functions; "PINN" represents the model in Fig. 3.

Table 1
Six different steady-state characteristics of Kresling origami structures.

Type	Mono-stable			Bi-stable				Special Mono-stable
	M1	M2	M3	B1		B2		P1
				1st	2nd	1st	2nd	
$\bar{h}$ at steady-state	>0	>0	$=0$	$=0$	>0	>0	>0	$=0$
$\bar{U}$ at steady-state	$=0$	>0	>0	>0	$=0$	$=0$	$=0$	$=0$

4. Results and discussions

In this section, we first validate the accuracy of potential energies predicted by the PINN through comparing them with those from Ref. [76] and by the exhaustive method. Second, the necessity of the advanced improvement in the PINN is proved, where two PINNs with or without the sorting operation are analyzed. Finally, two design cases are performed by the PINN, of which targets are respectively a desired potential energy curve and a set of desired potential energy points. In addition, the predictions and designs are conducted on the computing platform with an AMD Ryzen 9 7950X CPU.

4.1. Validation of the potential energy prediction

Different potential energy curves were provided in Ref. [76] to analyze six types of Kresling origami structures with different steady-state characteristics as described in Table 1. Additionally, the exhaustive method is an accurate method of calculating potential energy. Hence, to demonstrate the prediction reliability, we compare the potential energies by the PINN with those from Ref. [76] and by the

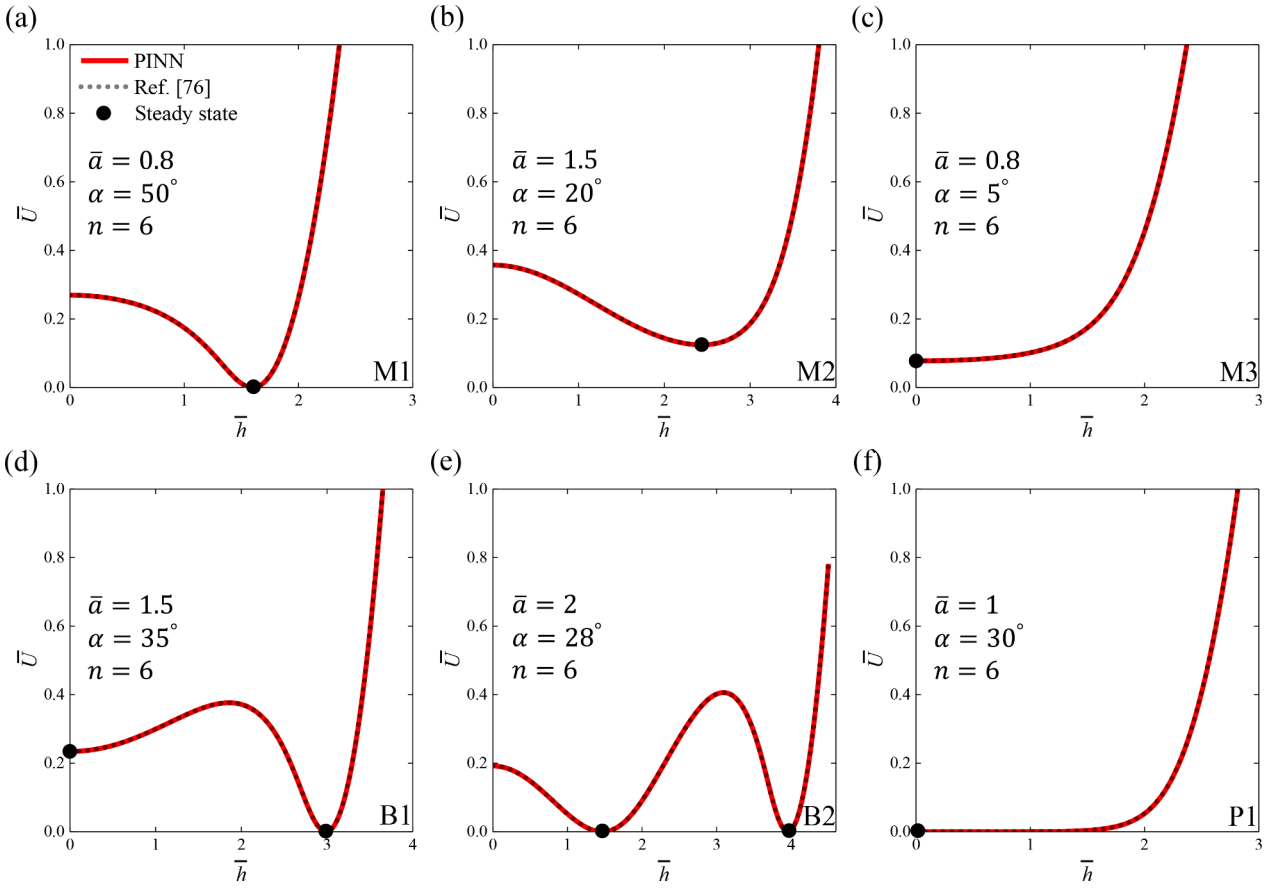


Fig. 6. Comparisons of potential energy curves by the PINN (red solid lines) and those from Ref. [76] (gray dotted lines); the black solid circles are the positions of steady states of Kresling origami structures and $n = 6$; M1, M2, and M3 represent three mono-stable structures; B1 and B2 represent two bi-stable structures; P1 represents a special mono-stable structure; (a) $\bar{a} = 0.8$ and $\alpha = 50^\circ$; (b) $\bar{a} = 1.5$ and $\alpha = 20^\circ$; (c) $\bar{a} = 0.8$ and $\alpha = 5^\circ$; (d) $\bar{a} = 1.5$ and $\alpha = 35^\circ$; (e) $\bar{a} = 2$ and $\alpha = 28^\circ$; (f) $\bar{a} = 1$ and $\alpha = 30^\circ$.

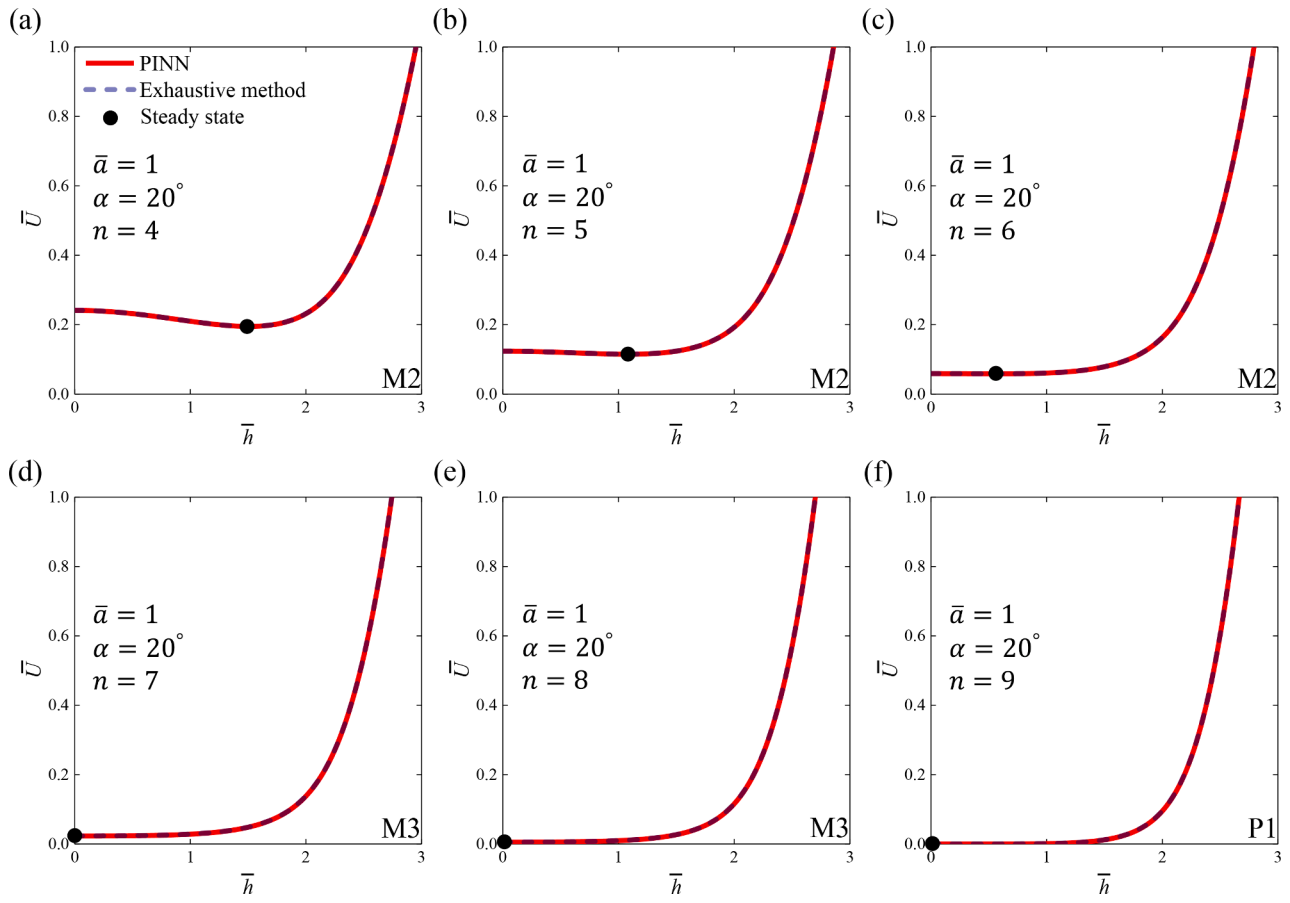


Fig. 7. Comparisons of potential energy curves by the PINN (red solid lines) and those by the exhaustive method (gray dashed lines); the black solid circles are the positions of steady states of Kresling origami structures and $\bar{a} = 1$ and $\alpha = 20°$; M2 and M3 represent two mono-stable structures; P1 is a special mono-stable structure; (a) $n = 4$; (b) $n = 5$; (c) $n = 6$; (d) $n = 7$; (e) $n = 8$; (f) $n = 9$.

Table 2
The credibility of results in Figs. 6 and 7.

Figure	The credibility L_G	
	PINN	Exhaustive method
6 (a)	1×10^{-26}	
6 (b)	2×10^{-24}	
6 (c)	1×10^{-25}	
6 (d)	3×10^{-25}	
6 (e)	2×10^{-24}	
6 (f)	1×10^{-25}	
7 (a)	2×10^{-25}	5×10^{-7}
7 (b)	3×10^{-25}	1×10^{-6}
7 (c)	1×10^{-25}	5×10^{-7}
7 (d)	2×10^{-25}	5×10^{-7}
7 (e)	3×10^{-25}	5×10^{-7}
7 (f)	2×10^{-25}	6×10^{-7}

exhaustive method. Eq. (9) is taken as the loss function of the PINN for potential energy prediction, where the dimension of the vector $\bar{h}$, namely m, is set to be 1000 in this section.

Fig. 6 shows the results by the PINN and those from Ref. [76], where the side number of the polygon n is 6, and other geometry parameters controlling the shape of Kresling origami structures are diverse. It can be seen that the predicted results by the PINN are perfectly consistent with those from Ref. [76]. Hence, the PINN can accurately predict the potential energy curves of Kresling origami structures with varying steady-state characteristics, which include three mono-stable structures (M1, M2, and M3), two bi-stable structures (B1 and B2), and a special mono-stable structure (P1). This comparison demonstrates the applicability of the PINN to all steady-state characteristics of Kresling origami structures. Fig. 7 considers six comparisons between the PINN and the exhaustive method under different n. Obviously, the predicted potential energy curves also align with those by the exhaustive method. This proves that the PINN can be applied to Kresling origami structures with different polygon side numbers. It can be also found that the steady-state type gradually transitions from M2, M3 to P1 as n increases. Hence, the PINN possesses the capability to accurately predict potential energy curves of Kresling origami structures with different steady-state characteristics and polygons.

L_G is used to assess the credibility of these results in Figs. 6 and 7, as shown in Table 2, where the closer L_G is to 0, the higher the credibility. The L_G values by the exhaustive method are all greater than 10^{-7}, and those by the PINN are all less than 10^{-23}. Although the obtained potential energy curves by the two methods in Fig. 7 are almost consistent, the PINN with higher credibility will be more beneficial to calculating higher-order physical quantities, such as force and stiffness. In addition, the training iterations of the PINN are 500 for each prediction, and the time costs 0.5 s. In the exhaustive method, the search grid dimensions of $\bar{h}$ and β are set to be 1000 and 3600, respectively, and the computational time is 40 s for each structure. Obviously, the PINN is more efficient and credible than the exhaustive method. Hence, the PINN will be a powerful tool in addressing forward and inverse challenges related to Kresling origami structures.

4.2. *The necessity of the advanced improvement*

To prove the necessity of the sorting operation in the last layer, two PINNs with or without the advanced improvement are compared. In this

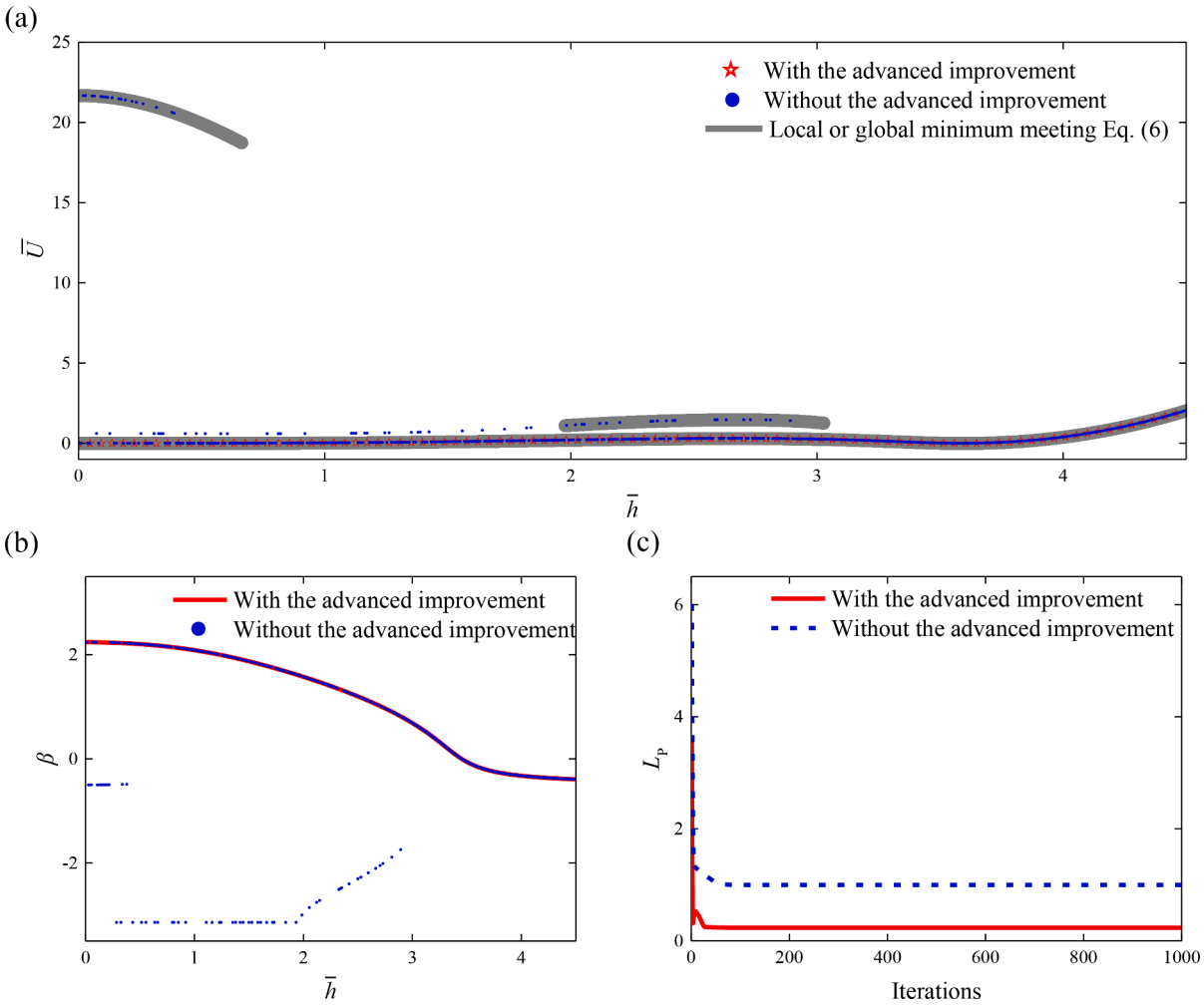


Fig. 8. Training results of the PINNs with or without the advanced improvement, where $\overline{h}$ is the height of the Kresling origami structure; (a) predicted values of the dimensionless potential energy $\overline{U}$ by the PINNs with the advanced improvement (the red pentagram) or without the advanced improvement (the blue circle), where the gray solid lines represent local or global minimum solutions meeting Eq. (6); (b) predicted values of the rotation angle of the middle polygon β by the PINNs with the advanced improvement (the red solid line) or without the advanced improvement (the blue circle); (c) learning curves of the PINNs with the advanced improvement (the red solid line) or without the advanced improvement (the blue dashed line), where L_P is the loss function for potential energy prediction, namely Eq. (9).

comparison, other hyperparameters of the PINN remain consistent, and the values of $\bar{a}$, α, n, and m are also kept the same, which are respectively 1.8, 30°, 6, and 1000.

As shown in Fig. 8(a), the results obtained with the advanced improvement perfectly align with the desired solutions which are the global minimum potential energies for each height, but those without the advanced improvement are unsatisfactory. In the latter situation, it is observed that although the majority of predicted values are desired solutions, there are still some results that deviate from the desired values, leading to the discretization of the potential energy curve. Fig. 8 (b) shows the output β by the two PINNs. Obviously, the β curve by the PINN without the advanced improvement is not continuous. The discontinuous potential energy points, which do not adhere to Eq. (6), are due to β values that are close to $-\pi$. In this situation, the differential coefficient of the activation function in the last layer is zero, making it difficult for the PINN without the advanced improvement to converge further. The potential energy points that match undesired local minima correspond to other β values that deviate from the continuous curve. These β values cause the PINN without the advanced improvement to become trapped in local optima. Fig. 8(c) shows the learning curves of the two PINNs. It is evident that it is the effect of the sorting operation that enables the predicted potential energies to align with the minimum total potential energy principle. Hence, the proposed advanced improvement is essential for the PINN, ensuring its ability to correctly predict potential energy curves of Kresling origami structures.

4.3. Inverse design

Since the nature of the PINN is a differentiable and continuous function, it can also realize the inverse design of Kresling origami structures by backpropagation. In the design process, the output potential energies are dependent variables of the PINN function, and the desired shape parameters, treated as independent variables (input of the PINN), can be obtained through gradient-based backpropagation method. The loss function for the inverse design is Eq. (10), which mandates the PINN simultaneously adheres to the target curve and the governing Eq. (6). Fig. 9(a) gives the designed results when n is set to be 5, 6, and 7, where m is set to be 1000. The designed curves with $n = 5$ and $n = 7$ are close to the target curve, and that with $n = 6$ is perfectly consistent with the target, where the time of each design costs 9.7 s after 10,000 iterations. Hence, the PINN can ensure that the potential energy curve of designed structures approximates a specified target curve. In addition, when $n = 7$, the slope of the potential energy curve between the second steady state and the maximum energy point is the lowest,

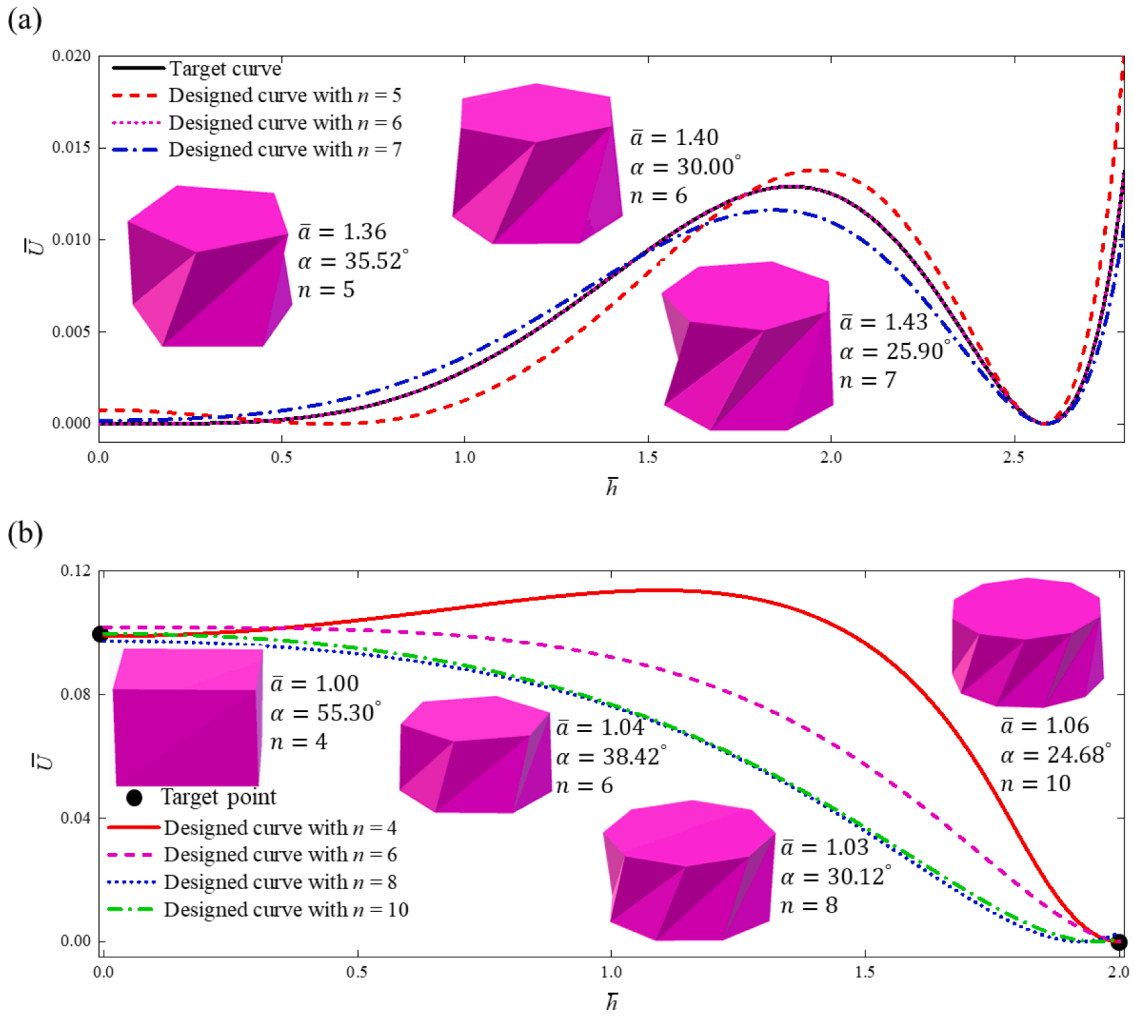


Fig. 9. Designed results by the PINN, where the dimensionless undeformed side $\bar{a}$ and the angle α between the sides a and b are design variables; $\bar{h}$ is the height of a Kresling origami structure; (a) meeting a specified target potential energy curve, where the black solid line is the target and other lines represent designed results when n is set to be 5, 6, and 7; (b) meeting two specified target points on the potential energy curve, where the black circles are targets and all lines represent designed results when n is set to be 4, 6, 8, and 10.

which means the force required to transition from the second steady state to the first steady state is minimal. Therefore, the designed structure with $n = 7$ may be a suitable choice for facilitating morphological transformations more easily. This structure also guarantees that the first steady state occurs at $h = 0$, distinguishing it from the designed structure with $n = 5$.

In some situations, the design only necessitates a few points on the potential energy curve to align with the target. As an example, it might be required that the potential energy value is zero at a specified non-zero $\bar{h}$, ensuring that the Kresling origami structure is in a stable state at this $\bar{h}$. Additionally, at $\bar{h} = 0$, a specified potential energy value is needed, enabling the Kresling origami structure to possess a predefined energy when absolutely folded. Fig. 9(b) shows the designed results in this situation when n is respectively 4, 6, 8, and 10, where the target energy is respectively set to be 0 or 0.1 at $\bar{h} = 0$ or 2, respectively. The loss function is the same with Eq. (10), where m is set to be 2. Obviously, the four designed potential energy curves satisfy the specified target points. The time cost for each design is 5.2 s after 10,000 iterations. Hence, the PINN has the capability to design Kresling origami structures according to a set of target potential energy points. Furthermore, the designed structure with $n = 4$ possesses two steady states where the first is at $h = 0$. Other structures only have one steady state around $h = 2$. If automatic expansion is a requirement for a folded structure, the designed results with $n = 6$, 8, and 10 are more suitable for this purpose.

It can be seen from the two cases that multiple different designs by the PINN can meet the same target, revealing the non-uniqueness nature of design problems. Meanwhile, these designed structures possess other different characteristics, making them suitable for different situations. Additionally, the PINN holds the potential to achieve multi-objective optimizations of Kresling origami structures if more target types are coupled into the PINN.

5. Conclusions

In this study, a novel PINN without labelled data is proposed to achieve the potential energy prediction and inverse design of Kresling origami structures. Since Kresling origami structures involve a transcendental equation, it becomes challenging for a traditional PINN to search for a desired solution. Therefore, an advanced improvement is coupled into the PINN to ensure the prediction correctness, which is the sorting operation in the last layer of the neural network. Given that the design variables must remain within valid ranges, two constraints are applied during the PINN design process. For the prediction and design problems, two loss functions with physical connotation are defined. To validate the reliability of the PINN, the potential energy curves predicted by the model are compared with those from Ref. [76] and by the exhaustive method. In the comparison, six Kresling origami structures

with different steady-state characteristics or with different n are taken into account. The predictions are perfectly consistent with the correct values, and only cost about one percent of the computational time of the exhaustive method. The PINN also demonstrates extremely high credibility in its predictions. Two PINNs with or without the sorting operation are analyzed, proving the effect and necessity of the advanced improvement in avoiding falling into local optima. The design capability of the proposed PINN is demonstrated through two design cases. The objectives of these cases are to match a target potential energy curve and a set of target potential energy points, respectively. The results show that the designed structures by the PINN can meet the expectations, and each design incurs a time cost of less than 10 s. This study opens up a novel avenue for addressing challenges related to origami structures, providing new insights into the AI-based origami structural design.

The PINN offers flexibility for origami structural design, allowing it to accommodate various forms of target potential energies by modifying the loss function. It also has the potential to design the bearing capacity or stiffness of Kresling origami structures, considering that force and stiffness are respectively the first and second derivatives of potential energy with respect to height. Furthermore, the PINN offers valuable references for foundation models, interpretable models, and uncertainty quantification in the field of AI-assisted origami.

In future work, more complex origami structures or metamaterials will be taken into account, and designs for specified requirements, such as quasi-zero stiffness, bandgap, and energy absorption, will be investigated. In addition, more powerful models based on physics-informed machine learning will be explored to solve challenges in the field of origami structures.

CRediT authorship contribution statement

Chen-Xu Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xinghao Wang:** Writing – review & editing, Validation, Investigation. **Weiming Liu:** Writing – review & editing, Supervision, Investigation. **Yi-Fan Yang:** Writing – review & editing, Visualization, Validation. **Gui-Lan Yu:** Writing – review & editing, Conceptualization. **Zhanli Liu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work is supported by the National Natural Science Foundation of China (No. 11972205, 11921002, and 11972210), the National Key Research Development Program of China (No. 2020-JCJQ-ZD-254), the National Key Research and Development Programs of China (No. 2022YFC3320502), and the National Postdoctoral Program for Innovative Talents (No. GZC20231310).

Appendix

The code in this study can be download in this link (https://bjtuedu cn-my.sharepoint.com/:f:/g/personal/19115031_bjtu_edu_cn/EiP 0Vi65csxGsf6mF794z1MBWEQ4cHQEW9dMqFzuouXUFg? e=kdAQAk).

References

[1] Chen H, Zhang XL, Zhang YY, Wang D, Bao DL, Que Y, Gao HJ. Atomically precise, custom-design origami graphene nanostructures. Science 2019;365(6457): 1036–40.
[2] Sareh P, Chen Y. Intrinsic non-flat-foldability of two-tile DDC surfaces composed of glide-reflected irregular quadrilaterals. Int J Mech Sci 2020;185:105881.
[3] Fonseca LM, Rodrigues GV, Savi MA. An overview of the mechanical description of origami-inspired systems and structures. Int J Mech Sci 2022;223:107316.
[4] Melancon D, Gorissen B, García-Mora CJ, Hoberman C, Bertoldi K. Multistable inflatable origami structures at the metre scale. Nature 2021;592(7855):545–50.
[5] Ji JC, Luo Q, Ye K. Vibration control based metamaterials and origami structures: a state-of-the-art review. Mech Syst Signal Process 2021;161:107945.
[6] Xi K, Chai S, Ma J, Chen Y. Multi-stability of the extensible origami structures. Adv Sci 2023;10(29):2303454.
[7] Morgan J, Magleby SP, Howell LL. An approach to designing origami-adapted aerospace mechanisms. J Mech Des 2016;138(5):052301.
[8] Troise A, Celli P, Cinefra M, Netti V, Buscicchio A. Reduced-order modelling of the deployment of a modified flasher origami for aerospace applications. Mater Res Proc 2023;37:547–52.
[9] Yang M, Defillion J, Scarpa F, Schenk M. Volume optimisation of multi-stable origami bellows for deployable space habitats. Acta Mech Solida Sin 2023;36(4): 1–17.
[10] Ze Q, Wu S, Nishikawa J, Dai J, Sun Y, Leanza S, Zhao RR. Soft robotic origami crawler. Sci Adv 2022;8(13):eabm7834.
[11] Yan W, Li S, Deguchi M, Zheng Z, Rus D, Mehta A. Origami-based integration of robots that sense, decide, and respond. Nat Commun 2023;14(1):1553.
[12] Kabutz H, Jayaram K. Design of CLARI: a miniature modular origami passive shape-morphing robot. Adv Intell Syst 2023;5(12):2300181.
[13] Beatini V, Gatheeshgar P, Rajanayagam H, Poologanathan K, Suntharalingam T, Perera D, Nagaratnam B. Integration of origami and deployable concept in volumetric modular units. Sci Rep 2022;12(1):19180.
[14] Liu B, Liao Y, Yang Y, Yang C, Tian Y, Yin H. Design and analysis of reconfigurable and deployable thin-walled architectural equipment inspired by Mirror-Miura origami patterns. Eng Struct 2023;286:116059.
[15] Megahed NA. Origami folding and its potential for architecture students. Des J 2017;20(2):279–97.
[16] Dey S, Fan C, Gothelf KV, Li J, Lin C, Liu L, Zhan P. DNA origami. Nat Rev Methods Primers 2021;1(1):13.
[17] Knappe GA, Wamhoff EC, Bathe M. Functionalizing DNA origami to investigate and interact with biological systems. Nat Rev Mater 2023;8(2):123–38.
[18] Kim M, Lee C, Jeon K, Lee JY, Kim YJ, Lee JG, Kim DN. Harnessing a paper-folding mechanism for reconfigurable DNA origami. Nature 2023;619(7968):78–86.
[19] Jianguo C, Xiaowei D, Ya Z, Jian F, Yongming T. Bistable behavior of the cylindrical origami structure with Kresling pattern. J Mech Des 2015;137(6): 061406.
[20] Zhang J, Zhang L, Wang C. Kresling origami-inspired reconfigurable antenna with spherical cap. Int J Mech Sci 2022;227:107470.
[21] Liu F, Terakawa T, Long S, Komori M. Rigid-foldable cylindrical origami with tunable mechanical behaviors. Sci Rep 2024;14(1):145.
[22] Chen Y, Feng H, Ma J, Peng R, You Z. Symmetric waterbomb origami. Proc R Soc A Math Phys Eng Sci 2016;472(2190):20150846.
[23] Rodrigues GV, Savi MA. Nonlinear dynamics and chaos of a waterbomb origami unit-cell considering different symmetry conditions. Mech Res Commun 2024;136: 104233.
[24] Fang H, Li S, Ji H, Wang KW. Dynamics of a bistable Miura-origami structure. Phys Rev E 2017;95(5):052211.
[25] Wang M, Karagiozova D, Lu G. Quasi-static three-point bending of sandwich panels with Miura-ori cores. Int J Mech Sci 2024;195:109010.
[26] Liu Y, Cai J, Feng J. Buckling suppression of a thin-walled Miura-origami patterned tube. PLoS ONE 2022;17(7):e0270228.
[27] Wan M, Yu K, Gu J, Zeng H, Sun H, Khatibi AA. 4D printed TMP origami metamaterials with programmable mechanical properties. Int J Mech Sci 2023; 250:108275.
[28] Lv W, Nie W, Zhang J, Wang Y, Guo S. Tapered origami tubes with non-planar cross sections. J Mech Robot 2024;16(7):074502.
[29] Zhai Z, Wu L, Jiang H. Mechanical metamaterials based on origami and kirigami. Appl Phys Rev 2021;8(4):041319.
[30] Fonseca LM, Rodrigues GV, Savi MA. An overview of the mechanical description of origami-inspired systems and structures. Int J Mech Sci 2022;223:107316.
[31] Meloni M, Cai J, Zhang Q, Sang-Hoon Lee D, Li M, Ma R, Feng J. Engineering origami: a comprehensive review of recent applications, design methods, and tools. Adv Sci 2021;8(13):2000636.
[32] Fan L, Liang J, Chen Y, Shi P, Feng X, Feng J, et al. Multi-stability of irregular four-fold origami structures. Int J Mech Sci 2024;268:108993.
[33] Ze Q, Wu S, Nishikawa J, Dai J, Sun Y, Leanza S, et al. Soft robotic origami crawler. Sci Adv 2022;8(13):eabm7834.
[34] Yan W, Li S, Deguchi M, Zheng Z, Rus D, Mehta A. Origami-based integration of robots that sense, decide, and respond. Nat Commun 2023;14(1):1553.
[35] Junfeng H, Guilin W, Jie L, Liang X, Xie YM. A modular continuous robot constructed by Miura-derived origami tubes. Int J Mech Sci 2024;261:108690.
[36] Ma J, You Z. Energy absorption of thin-walled square tubes with a prefolded origami pattern–Part I: geometry and numerical simulation. J Appl Mech 2014;81 (1):011003.
[37] Ye H, Liu Q, Cheng J, Li H, Jian B, Wang R, Ge Q. Multimaterial 3D printed self-locking thick-panel origami metamaterials. Nat Commun 2023;14(1):1607.

[38] Lam L, Chen W, Hao H, Li Z. Dynamic crushing and energy absorption of bio-inspired shear thickening fluid-filled origami metastructure. Eng Struct 2024;299:117122.
[39] Zhang X, Wang S, Durandet Y, Palanisamy S, Lu G. Energy absorption behavior of origami bellows under tension. Int J Mech Sci 2023;246:108143.
[40] Wang X, Qu H, Guo S. Tristable property and the high stiffness analysis of Kresling pattern origami. Int J Mech Sci 2023;256:108515.
[41] Mintchev S, Shintake J, Floreano D. Bioinspired dual-stiffness origami. Sci Robot 2018;3(20):eaau0275.
[42] Liu Q, Ye H, Cheng J, Li H, He X, Jian B, Ge Q. Stiffness-tunable origami structures via multimaterial three-dimensional printing. Acta Mech Solida Sin 2023;36(4):582–93.
[43] Filipov ET, Tachi T, Paulino GH. Origami tubes assembled into stiff, yet reconfigurable structures and metamaterials. Proc Natl Acad Sci 2015;112(40):12321–6.
[44] Li S, Fang H, Sadeghi S, Bhovad P, Wang KW. Architected origami materials: how folding creates sophisticated mechanical properties. Adv Mater 2019;31(5):1805282.
[45] Liu S, Peng G, Li Z, Li W, Sun L. Low-frequency vibration isolation via an elastic origami-inspired structure. Int J Mech Sci 2023;260:108622.
[46] Ishida S, Suzuki K, Shimosaka H. Design and experimental analysis of origami-inspired vibration isolator with quasi-zero-stiffness characteristic. J Vib Acoust 2017;139(5):051004.
[47] Ye K, Ji JC. An origami inspired quasi-zero stiffness vibration isolator using a novel truss-spring based stack Miura-ori structure. Mech Syst Signal Process 2022;165:108383.
[48] Liu W, Wu L, Sun J, Zhou J. Origami-inspired quasi-zero stiffness metamaterials for low-frequency multi-direction vibration isolation. Appl Phys Lett 2023;123(8).
[49] Liu C, Zhang W, Yu K, Liu T, Zheng Y. Quasi-zero-stiffness vibration isolation: designs, improvements and applications. Eng Struct 2024;301:117282.
[50] Han H, Sorokin V, Tang L, Cao D. Origami-based tunable mechanical memory metamaterial for vibration attenuation. Mech Syst Signal Process 2023;188:110033.
[51] Cai C, Zhou J, Wu L, Wang K, Xu D, Ouyang H. Design and numerical validation of quasi-zero-stiffness metamaterials for very low-frequency band gaps. Compos Struct 2020;236:111862.
[52] Lin Q, Zhou J, Wang K, Xu D, Wen G, Wang Q, Cai C. Low-frequency locally resonant band gap of the two-dimensional quasi-zero-stiffness metamaterials. Int J Mech Sci 2022;222:107230.
[53] Wang K, Zhou J, Ouyang H, Cheng L, Xu D. A semi-active metamaterial beam with electromagnetic quasi-zero-stiffness resonators for ultralow-frequency band gap tuning. Int J Mech Sci 2020;176:105548.
[54] Li Z, Wang K, Chen T, Cheng L, Xu D, Zhou J. Temperature controlled quasi-zero-stiffness metamaterial beam for broad-range low-frequency band tuning. Int J Mech Sci 2023;259:108593.
[55] Xu J, Jing J. Low-frequency band gaps in quasi-zero stiffness locally resonant metamaterial shaft. Int J Mech Sci 2024:108992.
[56] Dong HW, Zhao SD, Wang YS, Zhang C. Topology optimization of anisotropic broadband double-negative elastic metamaterials. J Mech Phys Solids 2017;105:54–80.
[57] Dong HW, Shen C, Zhao SD, Qiu W, Zheng H, Zhang C, Cheng L. Achromatic metasurfaces by dispersion customization for ultra-broadband acoustic beam engineering. Natl Sci Rev 2022;9(12):nwac030.
[58] Liu CX, Yu GL, Liu Z. Fast topology optimization of phononic crystal-based metastructures for vibration isolation by deep learning. Comput Aided Civ Infrastruct Eng 2023. https://doi.org/10.1111/mice.13100.
[59] Dang X, Feng F, Plucinsky P, James RD, Duan H, Wang J. Inverse design of deployable origami structures that approximate a general surface. Int J Solids Struct 2022;234:111224.
[60] Wang W, Liu K, Wu M, Li H, Lv P, Duan H. Instability-induced origami design by topology optimization. Acta Mech Solida Sin 2023;36(4):1–8.
[61] Ye Q, Gu XD, Chen S. Variational level set method for topology optimization of origami fold patterns. J Mech Des 2022;144(8):081702.
[62] Sareh P. The least symmetric crystallographic derivative of the developable double corrugation surface: computational design using underlying conic and cubic curves. Mater Des 2019;183:108128.
[63] Jiang T, Han Q, Li C. Design and bandgap optimization of multi-scale composite origami-inspired metamaterials. Int J Mech Sci 2023;248:108233.
[64] Chen Y, Yan J, Feng J, Sareh P. Particle swarm optimization-based metaheuristic design generation of non-trivial flat-foldable origami tessellations with degree-4 vertices. J Mech Des 2021;143(1):011703.
[65] Chen Y, Lu C, Yan J, Feng J, Sareh P. Intelligent computational design of scalene-faceted flat-foldable tessellations. J Comput Des Eng 2022;9(5):1765–74.
[66] Jordan MI, Mitchell TM. Machine learning: trends, perspectives, and prospects. Science 2015;349(6245):255–60.
[67] Rajkomar A, Dean J, Kohane I. Machine learning in medicine. N Engl J Med 2019;380(14):1347–58.
[68] Bender A, Schneider N, Segler M, Patrick Walters W, Engkvist O, Rodrigues T. Evaluation guidelines for machine learning tools in the chemical sciences. Nat Rev Chem 2022;6(6):428–42.
[69] Liu CX, Yu GL. Deep learning for the design of phononic crystals and elastic metamaterials. J Comput Des Eng 2023;10(2):602–14.
[70] Zhang D, Qin AK, Shen S, Trinchi A, Lu G. Energy absorption analysis of origami structures based on small number of samples using conditional GAN. Thin-Walled Struct 2023;188:110772.
[71] Zhu Y, Filipov ET. Harnessing interpretable machine learning for holistic inverse design of origami. Sci Rep 2022;12(1):19277.
[72] Moshtaghzadeh M, Bakhtiari A, Mardanpour P. Artificial Neural Network-based Finite Element method for assessing fatigue and stability of an origami-inspired structure. Eng Struct 2022;272:114965.
[73] Kaufmann J, Bhovad P, Li S. Harnessing the multistability of kresling origami for reconfigurable articulation in soft robotic arms. Soft Robot 2022;9(2):212–23.
[74] Han H, Tang L, Wu J, Sun S, Yin P, Cao D. Origami-inspired isolators with quasi-zero stiffness for coupled axial-torsional vibration. Aerosp Sci Technol 2023;140:108438.
[75] Dalaq AS, Daqaq MF. Experimentally-validated computational modeling and characterization of the quasi-static behavior of functional 3D-printed origami-inspired springs. Mater Des 2022;216:110541.
[76] Han H, Sorokin V, Tang L, Cao D. Lightweight origami isolators with deployable mechanism and quasi-zero-stiffness property. Aerosp Sci Technol 2022;121:107319.
[77] Yasuda H, Tachi T, Lee M, Yang J. Origami-based tunable truss structures for non-volatile mechanical memory operation. Nat Commun 2017;8(1):962.
[78] Zhai Z, Wang Y, Jiang H. Origami-inspired, on-demand deployable and collapsible mechanical metamaterials with tunable stiffness. Proc Natl Acad Sci 2018;115(9):2032–7.
[79] Yang X, Keten S. Multi-stability property of magneto-kresling truss structures. J Appl Mech 2021;88(9):091009.
[80] Ma W, Liu Z, Kudyshev ZA, Boltasseva A, Cai W, Liu Y. Deep learning for the design of photonic structures. Nat Photonics 2021;15(2):77–90.
[81] Karniadakis GE, Kevrekidis IG, Lu L, Perdikaris P, Wang S, Yang L. Physics-informed machine learning. Nat Rev Phys 2021;3(6):422–40.
[82] Samaniego E, Anitescu C, Goswami S, Nguyen-Thanh VM, Guo H, Hamdia K, Rabczuk T. An energy approach to the solution of partial differential equations in computational mechanics via machine learning: concepts, implementation and applications. Comput Methods Appl Mech Eng 2020;362:112790.